Why Mario Kart is Wrong!

An Analysis on Slipstream Dynamics

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GitHub Link: https://github.com/diegomperez25/Why_Mario_Kart_is_Wrong

1 Abstract

The overall goal of this project was to investigate slipstream dynamics in the physical world and compare them to the over-simplified mechanics of *Mario Kart World*. To do so, the first objective was to explore drafting ODEs based on previous research. This included creating phase portraits, contextualizing fixed points, checking for bifurcations, and analyzing variables. Once this was complete, relevant data was collected on *Mario Kart World* and its various characters/vehicles. This data was then used alongside the researched models to determine which character/vehicle combos are best/worst for drafting in the real world. The last objective was to compare the results of these simulations to in-game observations.

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2 Introduction

In 1996, *Mario Kart 64* introduced slipstreaming, or drafting, to the racing video game brand. This mechanism allows users who are trailing other racers to experience a slight boost after a certain amount of time. This boost is typically indicated by a wind-field at the front of the trailing vehicle. Since its first introduction, slipstreaming has been incorporated in most Mario Kart versions including the newest *Mario Kart World*.

Slipstreaming, however, is not just a video game mechanic. In fact, drafting was first observed by Junior Johnson in the 1960 Daytona 500. As a driver, he observed that following close behind another racer generated a slight increase in his vehicle's speed. Johnson then used drafting against his opponents' faster vehicles to ultimately win the race (Wikipedia contributors, 2024a) . Since then, researchers have investigated this phenomenon with all sorts of experiments. In the Journal of Mechanical Science and Technology, researchers built a computational model to investigate how slipstreaming between two vehicles affects their aerodynamic characteristics (Zabat et al., 1995). In the California PATH Research Report, Berkeley researchers used wind tunnels to explore the aerodynamic performance of multiple vehicles traveling in line (Jiang et al., 2001). In this report, I will be utilizing key findings from these projects to investigate drafting further and prove why *Mario Kart World* is ultimately wrong in its depiction of slipstreams.

A breakdown of this report's body is provided below:

(I): In Section 3.1, I use a traffic following model to demonstrate how *Mario Kart*'s slipstream mechanic affects driving behavior. I then investigate this ODE by identifying fixed points and explaining its long-term behavior.

(II): In Section 3.2, I introduce the real-world physics of slipstreaming and explain why it occurs. This provides the intuition for subsequent sections.

(III): In Section 3.3, I use the findings of previous research projects to construct a slipstream ODE that accurately captures the effect of different variables such as frontal surface area, mass, speed, and following distance on slipstream performance. Furthermore, I investigate the nature of this constructed ODE to determine/contextualize fixed points, identify bifurcations, and explore the long-term behavior of drafting.

(IV): In Section 3.4, I utilize collected data on *Mario Kart World* characters/vehicles to simulate how each combination would respond to slipstreaming effects. From these simulations, I observe and explain the 10 best/worst combinations for drafting. Lastly, we compare the results of these simulations to in-game observations.

2.1 Key Terms

The following key terms will be referenced frequently throughout this report. Their definitions have been provided below:

drag - the force acting opposite to the direction of the vehicle's motion with respect to the surrounding fluid (Wikipedia contributors, 2024b)

frontal area - the cross-sectional area of the vehicle perpendicular to the direction of the vehicle's motion (Fiveable, n.d.)

platooning - the method for driving a group of vehicles together at the same speed using electronic/mechanical coupling (Wikipedia contributors, 2024c)

wake - the region of recirculating flow behind a moving body caused by viscosity (Wikipedia contributors, 2024d)

slipstreaming/drafting - an aerodynamic technique where two moving objects are aligning in a close group to exploit the lead object's slipstream and thus reduce the overall effect of drag (SimScale, 2024)

2.2 Notation

Notation	Meaning
K	**(sensitivity/gain constant)** determines how aggressively a character tries to reach their optimal speed
$V(d(t))$	**(optimal velocity function)** defines the speed the game wants you to go based on your distance d from leader
$\dot{x}_f(t)$	**(current following velocity)** defines the current speed of the following character
λ	**(sensitivity to velocity difference constant)** represents how the character reacts to the relative speed between them and the leader
$\Delta v(t)$	**(velocity difference)** difference in speed between the leading and following cars
m	**(mass)** mass of the following racer

Notation	Meaning
$\frac{dv}{dt}$	**(acceleration)** rate of change of following kart's velocity over time
F_{engine}	**engine force)** forward force produced by the following kart's motor
ρ	**(air density constant)** represents thickness of air
A	**(frontal area)** represents cross-sectional area of character perpendicular to motion
$C_D(d)$	**(distance-dependent drag function)** represents defines drag coefficient for varying following distances
β	**(wake coefficient)** represents the scaling factor of a leading character's size on the length of their wake
α	**(strength coefficient)** scales the strength of the slipstream for all vehicles

2.3 Key ODEs

This report will mainly focus on the behavior of two ODEs. They are introduced below.

The Mario Kart ODE:
The first ODE we discuss is the model that illustrates Mario Kart's slipstream mechanism. It is based on the Full Velocity Difference Model from Jiang et al. (2001). The general form of the ODE is:

$$\ddot{x}_f(t) = K[V(d(t)) - \dot{x}_f(t)] + \lambda \Delta v(t) \tag{1}$$

The Real-World ODE:
The second ODE we discuss is the model that illustrates the true physical nature of drafting. It is based on *NASA*'s definition of drag force. $C_D(d)$ is derived from the observations of the Zabat et al. (1995):

$$\ddot{x}_f(t) = \frac{1}{m}\left(F_{engine} - \frac{1}{2}\rho A C_D(d)\dot{x}_f^2\right) \tag{2}$$

$$C_D(d) = C_\infty(1 - \alpha e^{-\beta d}) \tag{3}$$

3 Body

We now investigate the two ODEs above including the effect of different variables, fixed points, bifurcations, and long-term behavior.

3.1 How Mario Drafts

In *Mario Kart World*, slipstreaming is very simple. If a driver approaches another vehicle from behind and follows the leading vehicle for 1-2 seconds, the following kart experiences a set boost in speed that allows them to overtake the leader. Since this boost is constant and instantaneous, we do not need to take into consideration any aerodynamic features of the following vehicle to determine how it reacts to drafting. Instead, to define the effect of the slipstream, we rely on a car-following model presented by the University of Science and Technology of China (Jiang et al., 2001). This is the Full Velocity Difference Model (Eq. 1):

$$\ddot{x}_f(t) = K[V(d(t)) - \dot{x}_f(t)] + \lambda \Delta v(t)$$

It should be noted that this model is designed to represent two vehicles driving in line continuously. It assumes that the following vehicle wants to stay behind the leader, hence the second "braking" term.

We now analyze the variables of Eq. 1. The greater K is, the greater the following vehicle's acceleration will be. This makes sense intuitively since K represents how aggressively a character tries to reach their optimal speed. This is a direct reflection of the character's acceleration attribute. Additionally, the greater λ is, the lesser $\ddot{x}_f(t)$ is. This takes more thought to explain. Since λ represents the kart's sensitivity to the relative speed between them and the leader, it is essentially a reflection of how much braking power the following kart applies when they d becomes too small. We will use the kart's handling attribute to determine this quantity.

We now investigate the nature of $V(d(t))$. This function represents the speed that the game wants the following kart to reach given a specific following distance. Thanks to the simplicity of *Mario Kart*'s slipstream, we can define $V(d(t))$ more exactly. For any distance greater than the small distance required to slipstream, the following kart's "optimal" speed is the top speed reflected by the driver's speed attribute. Once the following distance reaches the threshold distance for drafting, $V(d(t))$ increases sharply by the set slipstream boost. Figure 1 provides a visual representation of $V(d(t))$.

Lastly, based on Eq. 1, we observe that $\ddot{x}_f(t)$ is dependent on $\dot{x}_f(t)$. This implies that when a following kart is trailing another vehicle at top speed outside the threshold distance, the "acceleration" they experience is dependent on the difference of speeds.

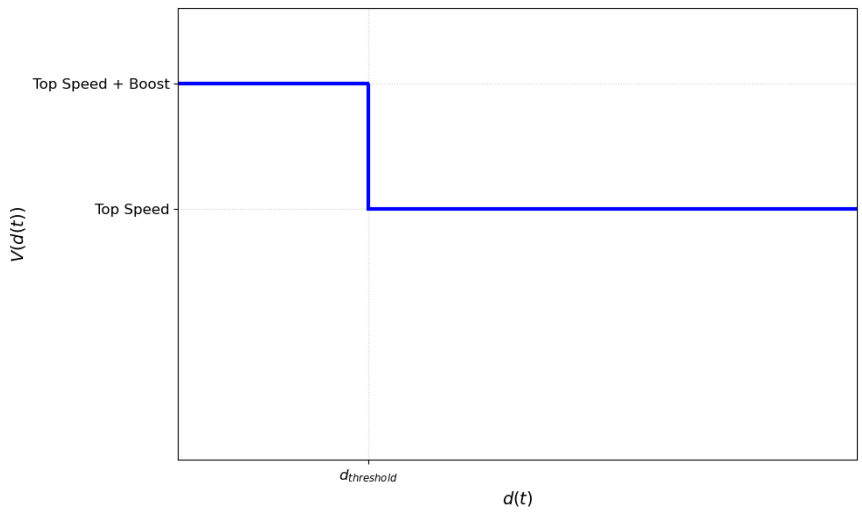

Figure 1: V(d(t))

Let us consider the model at the steady-state, when the following kart matches the leading kart's constant velocity $\dot{x}_L$ and $\ddot{x}_f = 0$. In this case,

$$\dot{x}_f = \dot{x}_L \Rightarrow \lambda \Delta v = \lambda(\dot{x}_L - \dot{x}_f) = 0$$

$$\ddot{x}_f = 0 = K[V(d^*) - \dot{x}_f] + 0 \Rightarrow V(d^*) = \dot{x}_f = \dot{x}_L$$

The distance d^* above is the ideal distance for drafting. We define a function for the general distance between two karts as follows:

$$d(t) = d^* + h(t)$$

where $h(t)$ above represents the perturbation from d^*. Since $d(t)$ is equal to the difference in positions of both the leading and following karts, we have:

$$d(t) = x_L(t) - x_f(t) = d^* + h(t)$$

By simple derivation, we have:

$$\ddot{d}(t) = \ddot{x}_L(t) - \ddot{x}_f(t) = \ddot{h}(t) \tag{4}$$

$$\dot{d}(t) = \dot{x}_L - \dot{x}_f = \Delta v = \dot{h}(t) \tag{5}$$

Since we are investigating the steady-state case, we also assume that $\ddot{x}_L(t) = 0$. This implies that $\ddot{h}(t) = -\ddot{x}_f(t)$.

We now look to construct a linear approximation for $V(d(t)$ using our new definition of $d(t)$. Using 1st order Taylor expansion, we have:

$$V(d) = V(d^* + h) \approx V(d^*) + V'(d^*)h \tag{6}$$

Substituting Eqs. 4, 5, and 6 into Eq. 1, we have:

$$-\ddot{h} = K[(V(d^*) + V'(d^*)h) - \dot{x}_f] + \lambda\dot{h} \tag{7}$$

But, we previously assumed that $\dot{x}_f = \dot{x}_L - \dot{h}$ in the steady state, so Eq. 7 can be simplified as below:

$$\begin{aligned} -\ddot{h} &= K[(V(d^*) + V'(d^*)h) - \dot{x}_L + \dot{h}] + \lambda\dot{h} && (8)\\ &= K[V(d^*) + V'(d^*)h - \dot{x}_L] + K\dot{h} + \lambda\dot{h} && (9)\\ &= K[V(d^*) + V'(d^*)h - \dot{x}_L] + (K+\lambda)\dot{h} && (10) \end{aligned}$$

Furthermore, since the following kart is at top speed at the steady state and its speed is equal to the speed of the leader, $V(d^*) = \dot{x}_f = \dot{x}_L$. Thus,

$$\begin{aligned} -\ddot{h} &= K[V(d^*) + V'(d^*)h - \dot{x}_L] + (K+\lambda)\dot{h} && (11)\\ &= K[\dot{x}_L + V'(d^*)h - \dot{x}_L] + (K+\lambda)\dot{h} && (12)\\ &= K[V'(d^*)h] + (K+\lambda)\dot{h} && (13) \end{aligned}$$

Adding $\ddot{h}$ to both sides, we have:

$$\ddot{h} + (K+\lambda)\dot{h} + K[V'(d^*)h] = 0 \tag{14}$$

It is insightful to note that Eq. 14 follows the general form of a damped harmonic oscillator: $\ddot{x} + 2\gamma\dot{x} + \omega_0^2 x = 0$. We now construct a system of 1st order ODES from Eq. 14 using the technique from this course.

$$\begin{aligned} \dot{h} &= y = f(h, y) && (15)\\ \dot{y} &= -(K+\lambda)y - K[V'(d^*)]h = g(h, y) && (16) \end{aligned}$$

By computing the Jacobian for this system, we have:

$$J = \begin{bmatrix} \frac{\partial f}{\partial h} & \frac{\partial f}{\partial y} \\ \frac{\partial g}{\partial h} & \frac{\partial g}{\partial y} \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -K[V'(d^*)] & -(K+\lambda) \end{bmatrix}$$

Computing the characteristic polynomial for J, we have:

$$P(\delta) = \det(J - \delta I) \tag{17}$$
$$= (-\delta)(-K - \lambda - \delta) + K[V'(d^*)] \tag{18}$$
$$= \delta K + \delta\lambda + \delta^2 + K[V'(d^*)] \tag{19}$$
$$= \delta^2 + \delta(K + \lambda) + K[V'(d^*)] \tag{20}$$

Solving for $P(\delta) = 0$, we use the quadratic formula to determine:

$$\delta = \frac{-(K + \lambda) \pm \sqrt{(K + \lambda)^2 - 4K[V'(d^*)]}}{2} \tag{21}$$

Now that we have computed the eigenvalues for J we can start to investigate the behavior of this system. Take note of the term outside the square root $-(K - \lambda)$. Assuming both K and λ are positive, this implies that the system is always stable. This means that no matter how much the following car may deviate from the ideal following distance, they will return to this distance through either braking or accelerating.

Looking at the system in isolation, we can determine its behavior for different scenarios. More specifically, we analyze the terms inside the square root.

- If $V'(d^*) > \frac{(K+\lambda)^2}{4K}$, the system is a stable spiral.
- If $V'(d^*) = \frac{(K+\lambda)^2}{4K}$, the system is a stable degenerate node.
- If $V'(d^*) < \frac{(K+\lambda)^2}{4K}$, the system is a stable node.

Now what happens when d^* approaches $d_{threshold}$? In this case, $V'(d^*)$ instantaneously goes to infinity. This breaks the stable system and allows the following racer to overtake the leader. This is the power of the slipstream effect in $MarioKart$. It allows drivers to escape the behavior of simply following the cars ahead and pushes them to pass their competitors. This instantaneous boost is not reflective of real-world slipstream dynamics, however. We discuss this more in the following sections.

3.2 Real Slipstreaming Explained

As mentioned before, drafting does exist in the real-world; however, it doesn't function exactly like it does in *Mario Kart*. Instead of an automatic boost that comes from following another vehicle, real drafting occurs when the following vehicle uses the reduced air resistance caused by the leading vehicle to gain speed. We explain this phenomenon more in depth in this section.

The key principle that allows drafting to occur is the "Boundary Layer Separation." This principle states that when airflow loses its ability to contour a moving vehicle, a low-pressure wake is created behind the vehicle, increasing drag resistance. When another vehicle enters this wake, they also experience the effects of this low-pressure region; however, since the region is in front, it pushes the following car forward. This is the process of slipstreaming that all racers utilize, from NASCAR to F1. Figure 2 illustrates how the air flow acting on a two-car system leads to this phenomenon (SimScale, 2024).

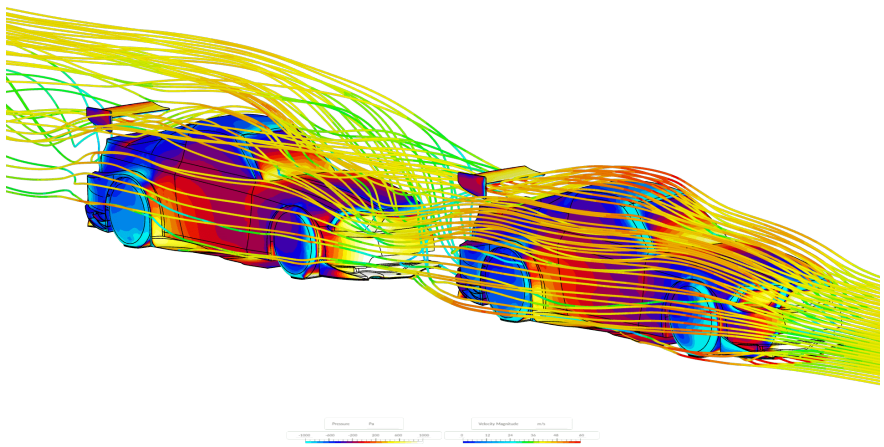

Figure 2: *Aerodynamic Drafting (Slipstreaming) in Racing*, 2024

As we can see in the diagram above, there are many factors to consider when determining how a following vehicle may react to drafting. The first prominent factor is the air itself. Denser air may find it easier to contour the leading vehicle, which can lead to a smaller wake that hinders the slipstream effect experienced by the following car. Additionally, the frontal area of both vehicles can affect this process. A larger frontal area for the leading car can generate a larger wake. A larger frontal area for the trailing car may not "fit" inside the wake produced. Finally, the following distance plays a key role in this interaction. If the trailing car is too far away from the leading car, they won't experience the full benefits of the slipstream.

3.3 The Real Slipstream Model

We now investigate Eq 2., which models the real-world slipstream mechanics of drafting:

$$\ddot{x}_f(t) = \frac{1}{m}\left(F_{engine} - \frac{1}{2}\rho A C_D(d)\dot{x}_f^2\right)$$

We will first analyze the variables of the model. The inverse relationship between $\ddot{x}_f(t)$ and m is given by Newton's 2nd Law, $F_{net} = ma = m\ddot{x}_f(t)$. The two terms inside the parentheses in the RHS of the equation represent the two forces acting on the following vehicle. These are the engine force that pushes the car forward and the drag force that holds the car back. Taking a closer look at the latter, we observe that both ρ and A have a positive relationship with the drag force. Additionally, the speed of the following car has a quadratic relationship with the drag force. This is interesting to note since it implies the faster the car is moving, the more air will push back against its motion.

The drag coefficient $C_D(d)$ requires more discussion than the rest of the variables. According to Hall (2021), C_d is usually determined experimentally, so we must rely on previous research to construct the drag coefficient function. Based on the graph of $C_{D,platoon}/C_{D,\infty}$ vs d (Figure 3) in the *California PATH Research Report*, we observe an exponential relationship between the following distance and the drag coefficient (Zabat et al., 1995). Although there is a hump in the ratio for very small distances, the relationship seems consistent for distances higher than 0.4 vehicle lengths. We will only consider $d > 0.4$ vehicle lengths for our model.

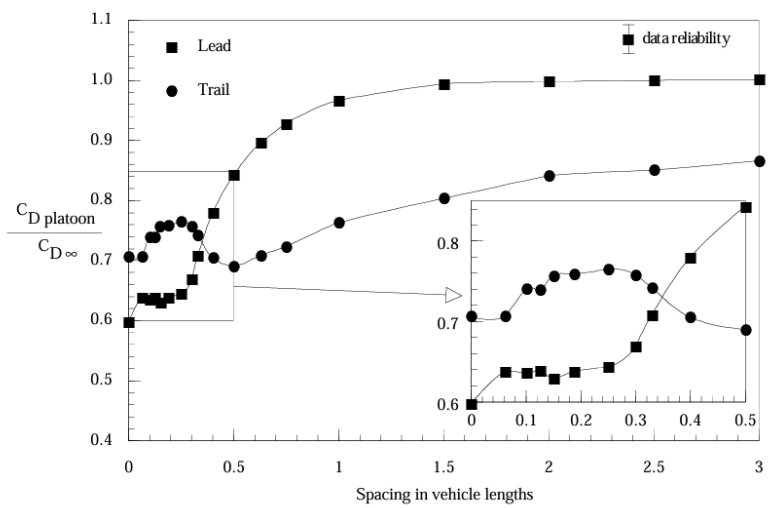

Figure 3: *California PATH Research Report, 1995*

Since the ratio approaches 1 as d grows larger, we define $C_D(d)$ as Eq 3:

$$C_D(d) = C_\infty(1 - \alpha e^{-\beta d})$$

Since we are basing this function on the findings of the graph above, we can determine an acceptable value for α. At $d = 0.4$, the $C_{D,platoon}/C_{D,\infty}$ ratio is 0.7. This implies that as d approaches zero, the drag coefficient drops by about 30 percent. Thus, $\alpha = 0.3$ is a good choice for our model.

As for β, this determines how fast C_D returns to C_D, ∞. Since this is essentially the size of the wake in the two-car system, β depends on the leading car. Since we will be assuming the same leading car for the rest of the report, we can use this graph to determine a good choice for β as well. Let us use the point on the graph above where $d = 1$ vehicle lengths. Since we are basing our model on $d > 0.4$, we substitute d for 0.6 to solve for β. Hence, we have:

$$\frac{C_D}{C_\infty} = 0.76 = 1 - \alpha e^{-0.6\beta} = 1 - 0.3e^{-0.6\beta}$$

Solving for β, we have $\beta \approx 0.24$. We will use this value for β for the duration of this report. Thus, our distance-dependent drag coefficient is defined as:

$$C_D(d) = C_\infty(1 - 0.3e^{-0.24d}) \tag{22}$$

We can now investigate the behavior of the slipstream ODE. Let us consider the case where $\ddot{x}_f(t) = 0$. In this case, $\dot{x}_f(t)$ is the terminal speed of the following vehicle. We solve for $\dot{x}_f(t)$ below:

$$\ddot{x}_f(t) = \frac{1}{m}\left(F_{engine} - \frac{1}{2}\rho A C_D(d)\dot{x}_f^2\right) = 0 \Rightarrow F_{engine} - \frac{1}{2}\rho A C_D(d)\dot{x}_f^2 = 0 \tag{23}$$

$$\Rightarrow v_t = \dot{x}_f = \sqrt{\frac{2F_{engine}}{\rho A C_\infty(1 - 0.3e^{-0.24d})}} \tag{24}$$

As we can see, $\dot{x}_f$ is a function of d. As d decreases, $\dot{x}_f$ increases. It is important to note that this increase in $\dot{x}_f$ is not instantaneous like the *Mario Kart* slipstream mechanism. Instead, $\dot{x}_f$ gradually changes with varying following distances. Furthermore, this implies that there is no "threshold distance." The slipstream effect is continuous, not spontaneous.

We now define a function for the speed of the following vehicle v based on v_t and u, a perturbation from v_t:

$$v = v_t + u$$

At v_t, $\dot{v} = 0$, which we showed above. Furthermore, at v_t, $F_{engine} = \frac{1}{2}\rho A C_D(d) v_t^2$, which can be easily seen from Eq. 23. We now substitute $\dot{v}$ and v into Eq. 2:

$$\dot{v} = \dot{u} = \frac{1}{m}\left(F_{engine} - \frac{1}{2}\rho A C_D(d)(v_t + u)^2\right) \tag{25}$$

$$= \frac{1}{m}\left(F_{engine} - \frac{1}{2}\rho A C_D(d)(v_t)^2 - \rho A C_D(d)(v_t)(u) - \mathcal{O}(u^2)\right) \tag{26}$$

$$\approx \frac{1}{m}\left(-\rho A C_D(d)(v_t)(u)\right) \tag{27}$$

Letting $\frac{\rho A C_D(d) v_t}{m} = A$, this simplifies to:

$$\dot{u} = -Au \tag{28}$$

This has the general solution:

$$u(t) = Ce^{-At} \tag{29}$$

As $t \to \infty$, we observe $u \to 0$. Hence, this system is stable. In terms of what this means physically, our results explain the long term behavior of the following vehicle's speed. If the following vehicle goes too fast, the drag force increase quadratically, pushing them back down to v_t. If the following vehicle goes slower than v_t, the drag force drops, allowing F_{engine} to accelerate the vehicle back to v_t.

We return back to Eq 2. to consider its phase portrait. To attain the general shape of this graph, we let $B = \frac{\rho A C_D(d)}{2m}$. Substituting this into Eq 2, we have:

$$\ddot{x}_f = \frac{F_{engine}}{m} - B\dot{x}_f^2 \tag{30}$$

Graphing $\ddot{x}_f$ versus $\dot{x}_f$, we have: As we can see, there is a stable fixed point for this system.

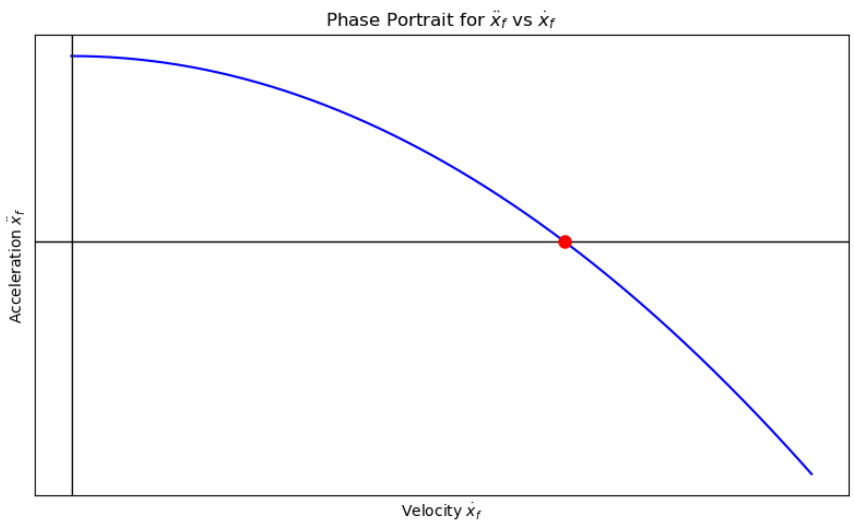

Figure 4: Phase Portrait for $\ddot{x}_f$ vs. $\dot{x}_f$

This confirms the previous conclusion that the following car tends to v_t as t approaches infinity. Analyzing Eq. 30, we observe that it follows the general form: $\dot{v} = h - rv^2$. We now graph the phase portrait for varying B values which correspond to varying r values.

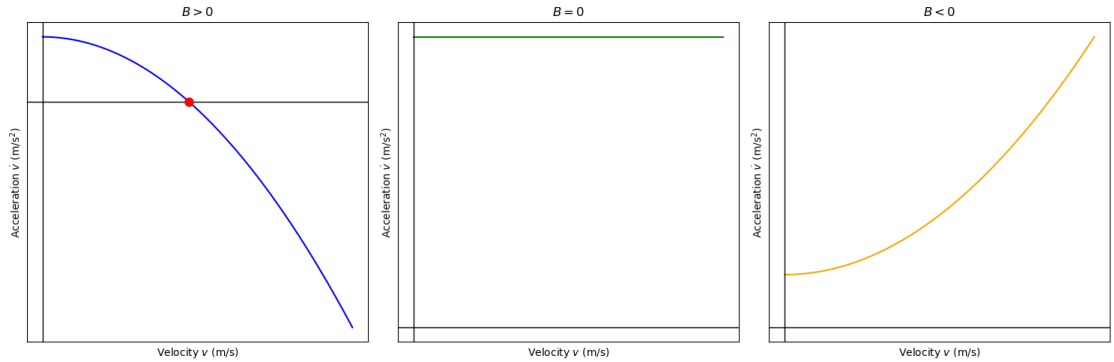

Figure 5: Phase Portraits for $B > 0$, $B = 0$, $B < 0$

As we can see above, as B goes from positive to negative, the fixed point illustrated in Figure 4 disappears. This implies that we have a bifurcation at $B = 0$. More specifically, since the fixed point is destroyed, we have a saddle-node bifurcation. Since $C_D(d)$ is the only variable in B that can reach 0, we now use our model to determine for what value d is B equal to zero. We solve for d below:

$$C_D(d) = C_\infty(1 - 0.3e^{-0.24d}) = 0 \tag{31}$$

$$\Rightarrow 1 - 0.3e^{-0.24d} = 0 \tag{32}$$

$$\Rightarrow d \approx -5.02 \tag{33}$$

Since $d < 0$ in this case, and we have constrained d to be greater than 0 in our discussion of slipstreaming, the bifurcation mentioned above has no physical significance, but it does add a mathematical boundary to keep in mind.

So, through our analysis of the slipstream ODE, we have found that the process of drafting is not as simple as *Mario Kart* paints it to be. At different distances, the drag force acting on the following car changes, affecting its achievable speed. If the car goes too fast, the drag force compensates for this by slowing the car down to its stable speed. Since we are more concerned with the boost generated by drafting, we focus more on the decreasing drag coefficient with decreasing distance as we simulate the slipstream performance of *Mario Kart* characters below.

3.4 How Mario Should Draft

In this section of our report, we will use the model from the previous section, as well as compiled data on *Mario Kart* characters/vehicles to determine which combos experience the best benefits from slipstreaming and which do not.

We start with the data collection on these characters/vehicles. For each, there are four attributes that are quantified by the game. These are 'Acceleration,' 'Handling,' 'Speed,' and 'Weight.' Using the data from Wheeler (n.d.), I compiled a dataframe with combined stats for each character/vehicle combo. Additionally, since frontal area is such an important variable for determining slipstream performance, I manually reviewed each character's design and assigned a 'Frontal Area' attribute based on their size. This process may be interpreted as subjective, but it was the best approach to generalize this feature. An example of the collected data is shown in Figure 6.

	Character	Vehicle	Acceleration	Handling	Speed	Weight	Frontal Area
0	Baby Daisy	Tune Thumper	18	15	3	4	3
1	Baby Daisy	Standard Bike	18	15	3	4	3
2	Baby Daisy	Cute Scoot	18	15	3	4	3
3	Baby Daisy	Rally Bike	17	15	4	4	3
4	Baby Daisy	Pipe Frame	17	14	4	5	3

Figure 6: This dataframe contains 1960 rows.

Taking this data, scaling attributes accordingly, and applying it to the model from the previous section, I measured the terminal velocity of each character/vehicle combo for varying following distances. Please note that I did set $\rho = 1.225$ and made C_∞ slightly dependent on weight to properly scale the model. I also used the 'Weight' attribute for m and both the 'Acceleration' and 'Speed' attributes to define F_{engine}. The leading vehicle is also assumed to be constant. An example of the results obtained is shown in Figure 7.

	Character	Vehicle	v_t @ d=inf	v_t @ d=1	v_t @ d=0.5	v_t @ d=0.25	v_t @ d=0.1	v_t_0.1 / v_t_inf	v_t_0.1 - v_t_inf
0	Baby Daisy	Tune Thumper	6.896504	7.890036	8.050140	8.141921	8.201328	1.189201	1.304824
1	Baby Daisy	Standard Bike	6.896504	7.890036	8.050140	8.141921	8.201328	1.189201	1.304824
2	Baby Daisy	Cute Scoot	6.896504	7.890036	8.050140	8.141921	8.201328	1.189201	1.304824
3	Baby Daisy	Rally Bike	7.739030	8.853940	9.033604	9.136597	9.203261	1.189201	1.464231
4	Baby Daisy	Pipe Frame	8.631369	9.874832	10.075212	10.190081	10.264431	1.189201	1.633062
5	Baby Daisy	Hyper Pipe	7.739030	8.853940	9.033604	9.136597	9.203261	1.189201	1.464231
6	Baby Daisy	Fin Twin	7.739030	8.853940	9.033604	9.136597	9.203261	1.189201	1.464231
7	Baby Daisy	Dolphin Dasher	7.739030	8.853940	9.033604	9.136597	9.203261	1.189201	1.464231
8	Baby Daisy	R.O.B. H.O.G.	8.394157	9.603446	9.798319	9.910031	9.982338	1.189201	1.588181
9	Baby Daisy	Mach Rocket	8.394157	9.603446	9.798319	9.910031	9.982338	1.189201	1.588181

Figure 7: v_t is depicted at different following differences d

Using this simulated data, I determined both the top 10 most optimal combos for slipstreaming and the top 10 least optimal combos. These were determined by sorting the dataframe by $v_{t,0.1}/v_{t,\infty}$ and $v_{t,0.1} - v_{t,\infty}$. In words, the top 10 most optimal combos experience the most optimal percentage increase and absolute increase in speed from drafting. The two figures below show the results of this sorting process.

	Character	Vehicle	v_t @ d=inf	v_t @ d=1	v_t @ d=0.5	v_t @ d=0.25	v_t @ d=0.1	v_t_0.1 / v_t_inf	v_t_0.1 - v_t_inf
0	Cow	Funky Dorrie	17.686037	20.233944	20.644532	20.879903	21.032250	1.189201	3.346212
1	Waluigi	Funky Dorrie	17.259805	19.746308	20.147000	20.376699	20.525374	1.189201	3.265569
2	Wiggler	Funky Dorrie	17.259805	19.746308	20.147000	20.376699	20.525374	1.189201	3.265569
3	Pianta	Rallygator	17.164910	19.637742	20.036231	20.264667	20.412525	1.189201	3.247615
4	Cow	Lobster Roller	16.653736	19.052927	19.439549	19.661182	19.804637	1.189201	3.150900
5	Chargin' Chuck	Blastronaut III	16.602789	18.994639	19.380078	19.601033	19.744049	1.189201	3.141261
6	Wario	Blastronaut III	16.602789	18.994639	19.380078	19.601033	19.744049	1.189201	3.141261
7	Pianta	Lobster Roller	16.449388	18.819140	19.201018	19.419931	19.561626	1.189201	3.112238
8	Stingby	Rallygator	16.413036	18.777550	19.158585	19.377014	19.518396	1.189201	3.105360
9	Stingby	Lobster Roller	16.413036	18.777550	19.158585	19.377014	19.518396	1.189201	3.105360

Figure 8: Top 10 most optimal combos for drafting.

	Character	Vehicle	v_t @ d=inf	v_t @ d=1	v_t @ d=0.5	v_t @ d=0.25	v_t @ d=0.1	v_t_0.1 / v_t_inf	v_t_0.1 - v_t_inf
0	Dry Bones	Tune Thumper	6.807513	7.788225	7.946263	8.036860	8.095500	1.189201	1.287987
1	Dry Bones	Standard Bike	6.807513	7.788225	7.946263	8.036860	8.095500	1.189201	1.287987
2	Dry Bones	Cute Scoot	6.807513	7.788225	7.946263	8.036860	8.095500	1.189201	1.287987
3	Peepa	Tune Thumper	6.807513	7.788225	7.946263	8.036860	8.095500	1.189201	1.287987
4	Peepa	Standard Bike	6.807513	7.788225	7.946263	8.036860	8.095500	1.189201	1.287987
5	Peepa	Cute Scoot	6.807513	7.788225	7.946263	8.036860	8.095500	1.189201	1.287987
6	Spike	Tune Thumper	6.807513	7.788225	7.946263	8.036860	8.095500	1.189201	1.287987
7	Spike	Standard Bike	6.807513	7.788225	7.946263	8.036860	8.095500	1.189201	1.287987
8	Spike	Cute Scoot	6.807513	7.788225	7.946263	8.036860	8.095500	1.189201	1.287987
9	Baby Daisy	Tune Thumper	6.896504	7.890036	8.050140	8.141921	8.201328	1.189201	1.304824

Figure 9: Top 10 least optimal combos for drafting.

Based on the tables above, Cow driving the Funky Dorrie is the best combo for real-world slipstreaming and Dry Bones driving the Tune Thumper is the worst. As for why this is, it is likely because the Cow combo has all-around balanced attributes. It has a medium frontal area, good speed, good acceleration, and a medium weight. On the other hand, Dry Bones has a low speed which implies it's engine force wouldn't aid much when entering a slipstream.

To visualize the behavior of both these sets of vehicles, the following figures depict the phase portraits for all vehicles in each class alongside each other.

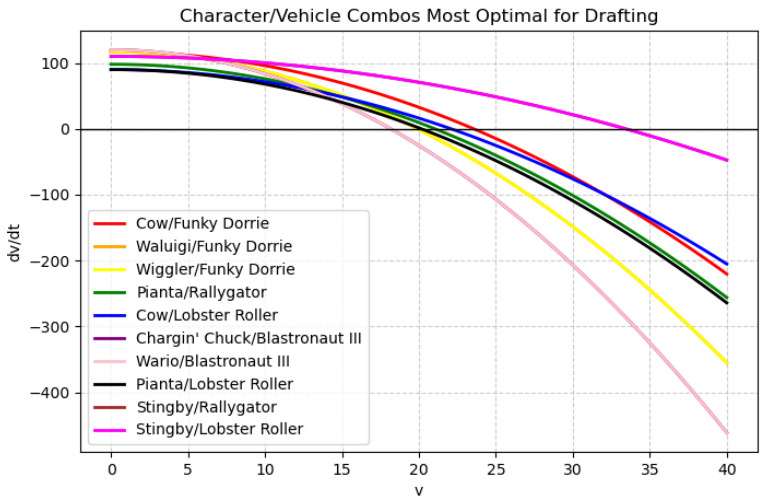

Figure 10: Phase Portraits for Top 10 Most Optimal.

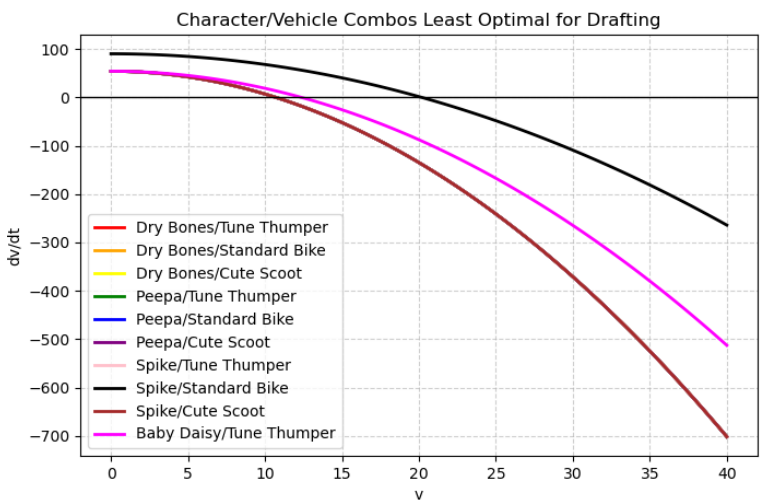

Figure 11: Phase Portraits for Top 10 Least Optimal.

What we observe from these graphs is that the most optimal combos generally tend to have higher speeds while drafting compared to the least optimal combos. It is also interesting to note that there is more variation in characters between the most optimal combos than the least optimal combos. This likely means that the character stats of the least optimal combos are what hinder their ability to capitalize on the benefits of slipstreaming.

Now that we have analyzed *Mario Kart*'s slipstream mechanic as well as real-world slipstream physics, and we have determined the character/vehicle combos most/least optimal for drafting, the last task we have is to compare our findings to what we observe while actually playing *Mario Kart World*. Unfortunately, in-game, there isn't much consideration for the size of the following characters nor their following distance when determining the slipstream effect. Thus, the above most optimal combos and least optimal combos don't have any merit when actually applied to the video game. Instead, because slipstreaming is just a standard boost, speed and acceleration are what players prioritize when building their combos. In fact, Toadette driving the Baby Blooper is allegedly the best combo in this respect according to the *Mario Kart World Statpedia* (ItsManu001, 2026). According to the simulations above, this combo is ranked 1257/1960 when it comes to slipstream performance. Thus, we can conclude that *Mario Kart World* is not exactly true to the physics of the real-world; however, that doesn't make it any less fun!

4 Acknowledgements

To help facilitate the construction of Figures 1, 4, 5, 10, and 11, I prompted Google Gemini to generate general code templates for my desired diagrams. I then customized these figures on my own including axis labeling, titling, and captioning.

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